Research Seminar «Data Science in Finance»

Modelling Regime Switching and Structural Breaks with an Infinite Dimension Markov Switching Model

Yong Song (2010), University of Toronto

Kokorev Artem

Regime switching or structural breaks?

Regime switching (Hamilton, 1989)

- Finite number of states (usually 2–4)
- Past states can recur — parameters are shared across visits
- Weakness: the future is assumed to look like the past; sudden new dynamics bias forecasts

Structural breaks (Koop & Potter, Chib, …)

- Infinite or very many states
- States never recur — every break brings brand-new parameters
- Weakness: data segments cannot share information, which costs estimation precision

Open questions: is U.S. inflation a regime-switching or a break process? Did real interest rates in 1981 get new dynamics, or return to a historical state? Existing models force you to pick one answer in advance.

Dirichlet process and stick-breaking

Dirichlet process $F \sim DP(\alpha_0, G_0)$

- A distribution over distributions: shape G_0 is the centre, concentration α_0 controls how close F is to G_0
- Draws are almost surely discrete, even if G_0 is continuous
- Posterior is again a DP: α_0 + n observations, G_0 mixed with the empirical distribution
- DP mixture (DPM): $g(y) = \Sigma\ p_i\ f(y \mid \theta_i)$ — an infinite mixture, so the number of components is not fixed

Stick-breaking representation

$$V_i \sim \text{Beta}(1, \alpha_0)$$

$$p_i = V_i \prod_{j<i} (1 - V_j)$$

$$\theta_i \sim G_0 \quad \text{(independent of p)}$$

Weights p and parameter values θ are generated independently — the paper exploits this to build two parallel hierarchies.

Problem for time series: a DPM treats observations as exchangeable, so it has no state persistence.

SDHDP-HMM: two parallel hierarchies

Specification

Hierarchy 1 — transitions

$$\pi_0 \sim \mathrm{SBP}(\gamma)$$

$$\pi_i \mid \pi_0 \sim \mathrm{DP}(c, (1-\rho)\pi_0 + \rho\delta_i)$$

Hierarchy 2 — data density

$$\lambda \sim G, \quad \theta_i \sim G_0(\lambda)$$

Dynamics

$$s_t \mid s_{t-1} = i \sim \pi_i$$

$$y_t \mid s_t = j \sim f(y_t \mid \theta_j, Y_{t-1})$$

Interpretation

- Sticky coefficient ρ puts extra weight on the diagonal of the transition matrix, so states persist
- Shared π_0 keeps the infinite transition matrix parsimonious: one stick-breaking vector instead of one per row
- $G_0(\lambda)$ is learned from all past states' parameters, so a newly born state starts from an informed prior
- Fox et al. (2008, FSJW) have only hierarchy 1; hierarchy 2 is the paper's addition

$E(P \mid \pi_0, \rho) = (1-\rho) \cdot$ [every row equals π_0] $+ \rho \cdot I$ — states can recur (regime switching) or be new (breaks).

Estimation and identification of breaks

Estimation (Bayesian MCMC)

- Each state: Gaussian AR(q); $G_0(\lambda)$ is normal-gamma with hyper-priors as in Pesaran et al. (2006)
- Finite approximation with L states (L = 10) and a block sampler: states by Chib's forward–backward smoother, π_0 and P by Dirichlet draws, θ by conjugacy, λ in blocks, ν by Metropolis–Hastings
- Label switching is avoided by reporting only label-invariant statistics, so no ordering constraints are needed
- Models are compared by out-of-sample log predictive likelihood (Kass–Raftery scale: > 5 is very strong)

Regime switching vs. break: the rule

- Look at one posterior path of states $s_1,\ldots,s_T$
- A state occupied in a single consecutive spell is non-recurrent; its start is a structural break
- A state that returns is recurrent; the start of each of its spells is a regime switching point
- Averaging over MCMC draws gives the probability of a break and of a switch at every date

Caveat from the paper: with finite data a true regime may look non-recurrent, so a switch can be labelled a break by accident.

Robust to misspecification, richer than both rivals

Log predictive likelihood, last 100 of 1000 obs.

True DGP \ Estimated	SDHDP-HMM	MS (3 states)	SB (4 states)
SDHDP-HMM	−170.55	−183.30	−264.65
Markov switching	−208.32	−208.10	−212.07
Structural break	−179.51	−187.26	−178.41

- If the truth is MS, SDHDP-HMM loses only 0.22 to the true model — not worth a mention
- If the truth is SB, it loses 1.1; the MS model loses 8.85 — very strong evidence against it
- If the truth is SDHDP-HMM, it beats MS by 12.75 and SB by about 94
- Estimated parameters and break/switch dates track the true ones closely

Reading the table: rows are the data-generating process, columns the estimated model. The true model is best in every row, but the SDHDP-HMM stays close to it when the truth is MS or SB, while the rivals fall far behind when the truth is the SDHDP-HMM.

U.S. real interest rates and inflation

Real interest rate, 1947Q1–2009Q4

- Regime switching dominates structural breaks: best MS (K = 5) scores −424.51, best SB (K = 15) −432.69
- SDHDP-HMM is best (−423.50), though not strongly better than MS; about 3.4 active states
- Supports Garcia–Perron (1973Q1 switch has probability 0.39) over Huizinga–Mishkin (Oct 1979 / Oct 1982: below 0.04)
- No sign of structural breaks in the posterior

Inflation, 1914–2010: log predictive likelihood

Model	Log pred. lik.
Linear AR(2)	−173.17
Structural break, K = 5	−98.69
Markov switching (Evans–Wachtel)	−92.50
FSJW (hierarchy 1 only)	−82.45
SDHDP-HMM	−74.07

Both features matter for inflation; the second hierarchy adds about 8 log points over FSJW. Posterior persistence in a new state is about 0.97.

Replication on U.S. inflation: setup

Data and models

- FRED CPIAUCNS, quarterly average; $\pi_t = 400 \cdot \Delta \log P_t$ (annualised %), 1913Q2–2026Q3, n = 454
- Train 1913–2000 (n = 351); test 2001Q1–2026Q3 (n = 103), covering 2008 and the 2020–23 spike
- Models: linear AR(2); MS-AR(2); SB-AR(2) (left-to-right); SDHDP-AR(2) (sticky finite HDP-HMM)
- One Gibbs sampler for all HMMs, L = 10, q = 2, 4 chains × 1500 iterations (burn 750, thin 2); only the transition prior differs ($\alpha = 3$, $\kappa = 10$, $\gamma = 3$)
- Score: out-of-sample log predictive likelihood, Kass–Raftery scale

Hypotheses

- H_1: recurrent regimes exist
- H_2: structural breaks exist
- H_3: a model with both mechanisms beats pure MS and pure SB out of sample

How it differs from the paper

- Quarterly data to 2026, not monthly to 2010
- Parameters fixed after training — no recursive re-estimation
- No second hierarchy $G_0(\lambda)$: emission priors are fixed, so the model is closer to FSJW

Out-of-sample results, test 2001Q1–2026Q3

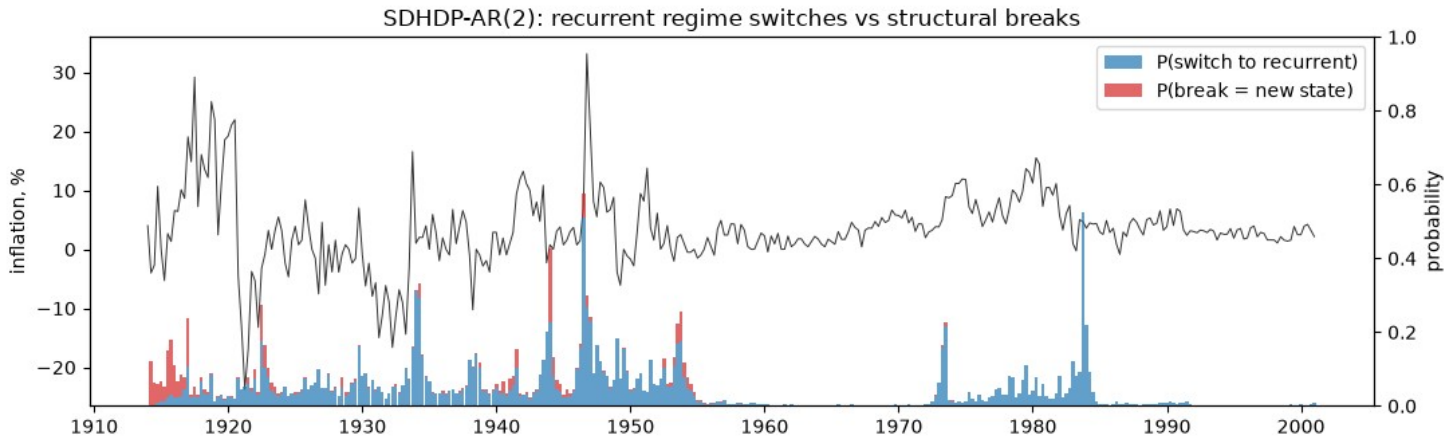
Model	Log pred. lik.	Δ vs AR(2)	Evidence vs AR(2)
SDHDP-AR(2)	−268.30	+7.34	very strong
MS-AR(2)	−269.85	+5.79	very strong
AR(2) OLS	−275.64	0	—
SB-AR(2)	−318.04	−42.40	very strong against

Verdict on the hypotheses

- H_1 supported: MS beats AR by 5.8
- H_2 rejected: SB loses to AR by 42.4 and to MS by 48.2; inflation is not a pure break process
- H_3 only weakly supported: SDHDP is best, but beats MS by just 1.55 (“positive”, not strong); it beats SB by 49.7

Same ordering as in the paper (SDHDP-HMM ≥ MS ≫ SB), but the margin over MS is far smaller than the paper reports, and it comes from a variant without the second hierarchy.

Where the sampler sees switches and breaks



SDHDP-AR(2), training sample: posterior probability of a switch to a recurrent state (blue) and of a break to a new state (red).

Active states (median per chain)
MS 9–10 of L = 10 · SB 2–4 · SDHDP 3–4

Most likely change dates

- 1946Q2: P(change) 0.58, mostly a switch
- 1983Q3: 0.53, all switch
- 1943Q4: 0.43, the largest break probability (0.20)

Break probabilities stay small throughout (at most 0.20): most detected changes are returns to earlier states. Links to WWII demobilisation and the Volcker disinflation are plausible readings, not tested.

Robustness: the ranking is not stable

Test start	n test	AR(2)	MS	SB	SDHDP	Best
2001	103	−275.64	−269.85	−318.04	−268.30	SDHDP
2011	63	−163.93	−159.87	−155.10	−158.06	SB
2020	27	−70.92	−67.99	−70.20	−71.52	MS

Reading

- SDHDP is second on the 2011 split but last on 2020 (below AR)
- Test windows of 63 and 27 quarters give noisy comparisons; extra splits used shorter chains (800 iterations)

Caveats

- Break/switch rule is a heuristic on posterior paths
- Pre-1945 NSA CPI is volatile; no convergence statistics beyond active-state counts
- Second hierarchy not implemented; β updated by a crude Dirichlet rule

Conclusion

1. One infinite-state Markov switching model nests regime switching and structural breaks; the number of states is estimated, not fixed

2. Two parallel hierarchies (transitions and data-density parameters) keep it parsimonious and improve forecasts

3. A label-invariant rule separates recurrent states (switches) from non-recurrent ones (breaks)

4. Real interest rates look like regime switching; inflation shows both features. Our replication on CPI to 2026 agrees on the ordering (SDHDP ≥ MS ≫ SB), with a small margin over MS

5. Caveats: the break/switch rule is a heuristic; evidence is predictive likelihood only; in our replication the ranking flips across test windows and the second hierarchy is not implemented

Research Seminar «Data Science in Finance»

Modelling Regime Switching and Structural Breaks with an Infinite Dimension Markov Switching Model

Yong Song (2010), University of Toronto

Kokorev Artem